

Image Denoising: How the Field Actually Evolved

Not ten papers in order. One technical story: what each method could not do, what the next one changed, and which line of code changed to do it.

THE QUESTION

Why did each innovation
have to be invented?

ANCHOR PAPER

DnCNN — Zhang et al.
IEEE TIP 26(7), 2017

PRESENTERS

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REFERENCE

100-slide master deck
submitted separately

Four questions we will answer about every method

Each algorithm gets the same four-part treatment, so the comparison between them is structural rather than anecdotal. The methods that were state of the art get a fifth question.

1

What could the previous method not do?

Every innovation in this field was a response to a *named, measured* failure — usually named by the paper that fixed it, which is what makes this timeline evidenced rather than imposed.

2

What changed technically?

The algorithmic change: a different prior, a different training target, a different operator. Stated precisely enough that you could implement it.

3

Which line of code changed?

A Before → After from the reference implementations. Contribution → algorithmic change → **implementation change** → measured improvement, as one chain.

4

What did it buy, and what did it cost?

The measured improvement, the applications it opened, and the new limitation it created — because every step in this story created the next problem.

And for the SOTA methods, a fifth: what made it state of the art, *on which benchmark and in which years*, and why it did not make the others obsolete.

One equation, three unknowns, eighteen years

Given only y , infinitely many splits into $x + v$ are arithmetically valid. Every method in this talk supplies the missing information in a different way — that supplied information is called the prior, and the whole story is about where it lives.

$$y = x + v$$

y — what the sensor produced · x — the clean image we want · v — the noise

- 01 **The classical answer** writes the prior down as a formula: minimise $\frac{1}{2}\|y - x\|^2 + \lambda \cdot \Phi(x)$. You can read Φ , differentiate it, and argue about it.
- 02 **The deep answer** replaces Φ with a tensor of learned weights. You cannot read it, but you can download it and it runs in 60 ms.
- 03 **The self-supervised answer** takes the prior partly from the test image itself, so no clean data is needed at all.
- 04 **Where the prior lives, and who writes it, is the single axis this whole talk runs along.**

What 'state of the art' means in this talk

The phrase is used loosely in most presentations. We use it with three qualifiers attached, because that is what makes the SOTA comparison later in the talk meaningful rather than rhetorical.

A SOTA claim needs three things

- **A benchmark and a metric.** 'Best denoiser' is meaningless. 'Highest PSNR on BSD68 at $\sigma=25$ ' is a claim you can check.
- **A comparison protocol.** Two papers reporting on the same dataset can still disagree, so wherever a row compares two methods we take both numbers from **one table**.
- **Honesty about what is not reported.** No paper here states how long it held a position, so this talk does not claim SOTA *windows* — only the publication date and which table each method leads.

Which is why there is no single winner

- Different benchmarks measure different capabilities: BSD68 measures average-case Gaussian removal, Urban100 measures **long-range self-similarity**, SIDD measures **real sensor noise**.
- A method can lead one and trail another. Restormer, NAFNet and MambaIR each lead a different table *today*.
- **We return to this on slide 36** with the numbers, because 'why are there several SOTAs' is the question this talk is built to answer.

reading, grounded in the benchmark definitions used by DnCNN17, Restormer22, NAFNet22

1

The Hand- Written Prior

— BASELINE · 2007 – 2014

— In this part

01 BM3D — the baseline everything is measured against

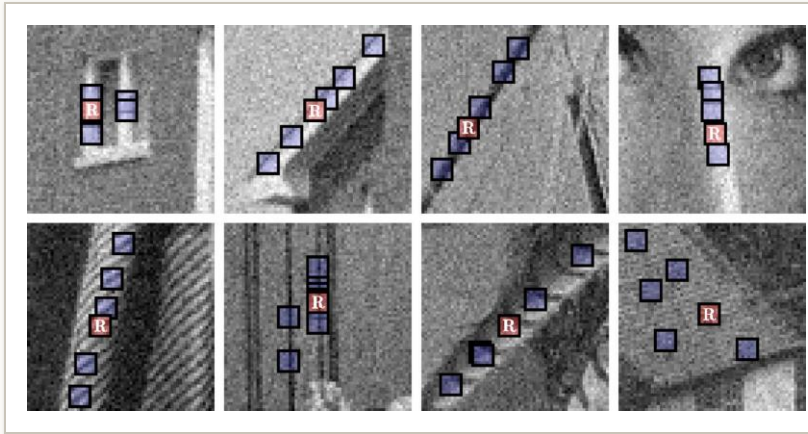
02 WNNM — the classical ceiling

03 Bottleneck 1 — why the era ended

Where the field started: a formula a human wrote, re-solved from scratch for every image.

The innovation: filter patches together, not alone

Dabov et al., IEEE TIP 2007. Previous transform-domain methods shrank each patch independently. BM3D's innovation is to *group* similar patches first, creating a third axis along which the signal is nearly constant and the noise is not.



Block matching in practice: the reference patch *R* and the similar patches BM3D groups with it before collaborative 3-D filtering. [BM3D07 · Fig. 2]

What changed

Block-match similar patches into a 3-D stack, transform the **stack**, hard-threshold, invert, aggregate — then repeat using the cleaner estimate as a Wiener pilot.

Why it works

The paper's own ablation: even a transform that is *random apart from the DC* loses only 0.1–0.4 dB. **The grouping does the work, not the transform.** [BM3D07 §V]

The innovation: filter patches together, not alone

Impact

BSD68 $\sigma=25$: **28.57 dB** at 2.85 s per 512×512. Became the reference line every paper for a decade had to beat.

Still in use

Appears as a live baseline in Noise2Noise (2018), Noise2Void (2019), ZS-N2N (2023) and Restormer (2022) — a seventeen-year-old algorithm is still what new zero-shot methods must beat, because it needs no training data.

Applications

The zero-shot comparison in those same papers, and the fallback where there is neither training data nor a GPU. On real sensor noise it still beats an AWGN-trained DnCNN by 1.99 dB. [[Restormer22 Table 6](#)]

BM3D07 §II-C, §V · DnCNN17 Table II, Table IV · baselines in N2N18, N2V19, ZS-N2N23

WNNM vs BM3D — the same grouping, a better estimator

Gu et al., CVPR 2014. WNNM keeps BM3D's non-local grouping and replaces what happens to the group. This is the first clean Before → After in the story, and it is entirely inside one step.

ASPECT	BM3D (2007)	WNNM (2014)	WHY IT MATTERS
Core algorithm	Group patches, shrink in a 3-D transform	Group patches, recover a low-rank matrix	Stacked near-copies are low rank by construction — a stronger structural assumption
Key innovation	Collaborative hard-thresholding	Weighted nuclear norm: $w_i \propto 1/\sigma_i$	Large singular values carry the image and are shrunk <i>less</i> ; standard NNM shrinks all equally
Implementation change	3-D DCT + threshold	SVD per patch group + per-value shrinkage, iterated	One transform call becomes an iterated SVD — the entire cost difference
Performance	28.57 dB · 2.85 s	28.83 dB · 773.2 s	+0.26 dB for a 271× slowdown
Limitation	Fixed hand-set parameters	Non-convex; still per-image optimisation	Neither can reuse anything learned from one image on the next
Applications	General-purpose, real-time-ish	Offline / quality-critical restoration	The cost makes WNNM impractical for interactive or batch use

This row is the whole argument for what comes next. Seven years of hand-designed priors bought 0.26 dB on BSD68 at $\sigma=25$, and the best method became 271× slower than the second best. The era had hit a wall in quality *and* cost simultaneously.

DnCNN17 Table II, Table IV · WNNM14 §3, Theorem 2



Why the era ended — stated by the paper that ended it

DnCNN's introduction names the classical era's drawbacks in its own words. Quoting the paper that broke the bottleneck, rather than asserting it ourselves, is what makes this an evidenced timeline.

DRAWBACK 1

The prior is paid for once per image

Prior-based methods *involve a complex optimization problem in the testing stage*. Nothing learned on image 1 helps image 2. WNNM: 773 s, every time. [\[DnCNN17 §I\]](#)

DRAWBACK 3

Its form limits the ceiling

Performance is *inherently restricted to the specified forms of prior*. You cannot exceed what your formula can express, however well you tune it. [\[DnCNN17 §I\]](#)

DRAWBACK 2

The prior is written by hand

The models are *generally non-convex and involve several manually chosen parameters* — block size, thresholds, λ , different values above and below $\sigma=40$. [\[DnCNN17 §I\]](#)

DRAWBACK 4

One model per noise level

Bridge methods *train a specific model for a certain noise level and are limited in blind image denoising*. In a real camera, σ is never known. [\[DnCNN17 §I\]](#)

DnCNN17 §I — all four quoted from the paper's introduction

2

The Prior Becomes a File

— THE BREAK · 2016 – 2017

— In this part

01 TNRD — the bridge nobody skips

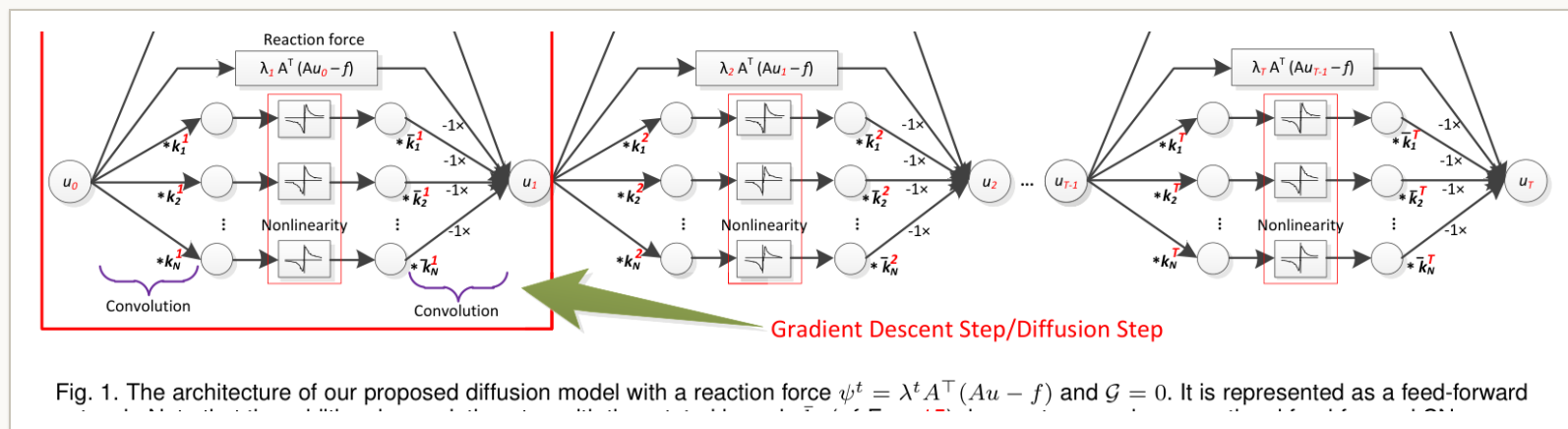
02 DnCNN — innovation, implementation, impact

03 Bottlenecks 2 and 3

One architecture answers all four drawbacks at once — and creates the two problems the rest of the talk is about.

Unrolling — the step that made the transition continuous

Chen & Pock, TPAMI. Worth ninety seconds because it is the conceptual bridge: it shows that a classical optimiser and a neural network are the same object viewed differently, which is why the move to deep learning was not a rupture.



The diffusion network unrolled as a feed-forward graph: learned filters, trainable nonlinearity and reaction force applied at every stage. [TNRD16/17 · Fig. 1]

ASPECT	CLASSICAL OPTIMISER	TNRD (UNROLLED)	WHY IT MATTERS
Iteration count	Run to convergence, unbounded	Truncated to a fixed K (usually < 10)	Runtime becomes a constant — the first half of bottleneck 1 is gone
Parameters	Same hand-set rule every iteration	Every stage has its own learned filters and nonlinearities	The prior is no longer one formula; it is K different learned ones
What is learned	Nothing	Filters <i>and</i> influence functions, jointly, from data	Earlier diffusion models fixed the filters and chose the diffusivity
Implementation	<code>while</code> loop to a tolerance	A fixed stack trained by back-propagation	Structurally a CNN — and <i>well-suited for parallel computation on GPUs</i>

Unrolling — the step that made the transition continuous

ASPECT	CLASSICAL OPTIMISER	TNRD (UNROLLED)	WHY IT MATTERS
Performance	WNNM 28.83 dB · 773 s	28.92 dB · 0.032 s (GPU)	Best quality-per-second of the era
Limitation	—	Still one model per noise level	Blind denoising remains unsolved

Keep this in mind for the next slide. DnCNN derives part of its own justification from a one-stage TNRD — the two are directly connected, not merely adjacent in time.

DnCNN vs TNRD — remove the optimiser entirely

Zhang et al., IEEE TIP. TNRD unrolled the optimiser; DnCNN throws away the optimisation heritage and keeps only the network. The technical change is small, the consequence is the largest single step in this story.

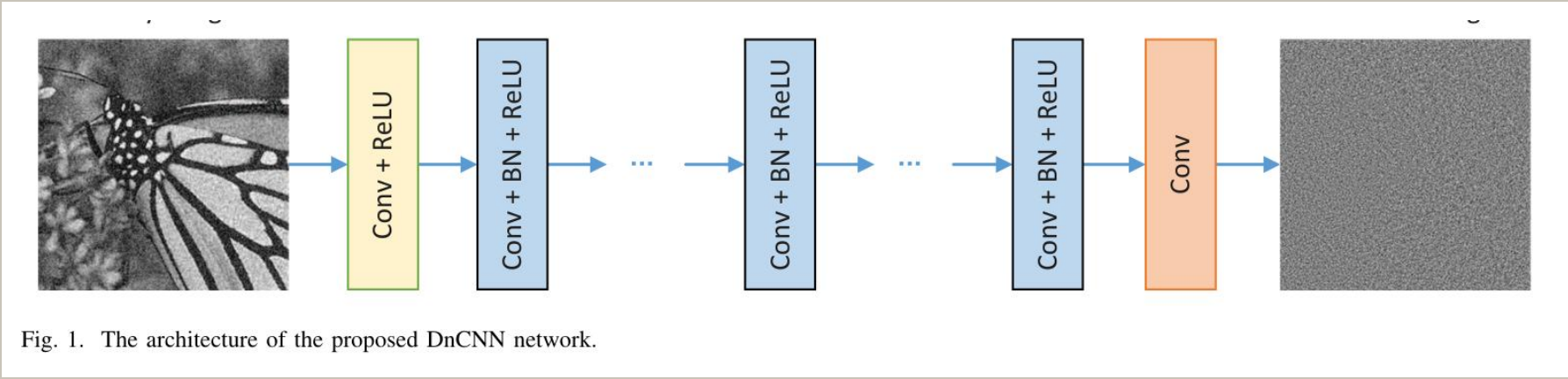


Fig. 1. The architecture of the proposed DnCNN network.

The DnCNN architecture: Conv(+BN)+ReLU layers predict the residual noise, not the clean image. [DnCNN17 · Fig. 1]

ASPECT	TNRD (2016)	DNCNN (2017)	WHY IT MATTERS
Core algorithm	K diffusion stages, PDE-derived	Plain deep CNN: Conv-BN-ReLU ×17, no pooling	The architecture no longer has to resemble an optimiser at all
What the net outputs	The restored image	The noise — $x = y - R(y)$	Residual learning: the 'do nothing' solution moves to zero, which is trivially reachable
Key enabler	Learned influence functions	Residual learning + batch normalisation, together	Neither works nearly as well alone — the paper's most-cited empirical finding
Noise levels	One model per σ	DnCNN-B: one blind model, $\sigma \in [0, 55]$	Drawback 4 solved; no noise estimate needed at test time

DnCNN17 §II-A, §III-A, §III-C, §III-D, Table II, Table IV · TNRD16

DnCNN vs TNRD — remove the optimiser entirely

ASPECT	TNRD (2016)	DNCNN (2017)	WHY IT MATTERS
Performance	28.92 dB · 0.032 s	29.23 dB · 0.060 s (blind: 29.16)	+0.6 dB over BM3D at <i>all three</i> noise levels
Applications	Gaussian denoising	One model for denoising + super-resolution + JPEG deblocking	Redefine v and the same network solves three tasks

DnCNN17 §II-A, §III-A, §III-C, §III-D, Table II, Table IV · TNRD16

The innovation is eight characters

`cszn/KAIR, models/network_dncnn.py`, by DnCNN's own first author. The architecture is a list comprehension; the contribution is the return statement.

KAIR/models/network_dncnn.py — the whole network

```
m1_head = B.conv(in_nc, nc, mode='C'+act_mode[-1])
m2_body = [B.conv(nc, nc, mode='C'+act_mode)
3         for _ in range(nb-2)]          # ← the entire network
m4_tail = B.conv(nc, out_nc, mode='C')
5
def forward(self, x):
7     n = self.model(x)
8     return x-n                          # ← the contribution
9
#0 BEFORE (discriminative CNNs, ~2012-2016):
#1     return self.model(x)              -> learn  $F(y) = x$ 
#2 AFTER (DnCNN, 2017):
#3     return x - self.model(x)          -> learn  $R(y) = v$ 
```

- 01 **Why that one change matters.** From ResNet: a mapping close to the identity is hard to optimise. And y is much more like x than it is like v — so learning the *image* forces the network toward an identity transform, which seventeen ReLU layers have no natural way to express.
- 02 **Learning the noise moves the trivial solution to zero**, reachable by driving weights toward zero. Same function class, far easier optimisation landscape.
- 03 **The synergy, in one sentence.** Residual learning strips the image out of the hidden layers, making activations Gaussian-like and content-independent — which is exactly the regime batch-norm's batch statistics assume. *Without* residual learning, BN *has a certain adverse effect on convergence*.
- 04 **Where the prior went.** Nowhere in this file. It is in the `.pth` checkpoint this class will load. That is the moment the prior stopped being readable — and started being downloadable.

measured — cszn/KAIR, verbatim · DnCNN17 §II-A, §III-C

What made it state of the art, and for how long

A SOTA claim with its qualifiers attached: the benchmark, and the margin measured against the ceiling the literature had already established.

+0.6 dB

OVER BM3D, ALL THREE Σ

On BSD68. Prior work put the practical ceiling over BM3D at **0.3 dB** and the estimated theoretical bound at **0.7 dB** — so DnCNN took about **85% of the available headroom**. [DnCNN17 §IV-B]

12 900×

FASTER THAN WNNM AT 512²

773.2 s CPU → 0.060 s GPU. The device differs and the paper says so; the order of magnitude is the point. [DnCNN17 Table IV]

3 tasks

ONE MODEL, ONE LOSS

DnCNN-3 handles denoising, super-resolution and JPEG deblocking by redefining v , beating a specialist baseline on all three. [DnCNN17 §III-D, Table V]

- 01 **And it did it blind.** DnCNN-B reaches 29.16 dB without being told σ , beating every noise-level-specific competitor in the table. That is drawback 4 closed.
- 02 **Applications it opened:** real-time denoising in imaging pipelines; the `.pth` checkpoint as a shippable artefact; and denoising as a *reusable component* — a trained denoiser can be dropped into variable-splitting solvers as a general-purpose prior. [SCUNet23 §1]
- 03 **What it was not.** On *real* sensor noise DnCNN was never the leader — it scores 23.66 dB on SIDD. That gap is slide 21's subject.

DnCNN17 §IV-B, Table II, Table IV · SCUNet23 §1

What DnCNN broke, and the two bills it left

This slide is the hinge of the whole talk. Everything after it is the field paying off these two debts — and they were paid off by two different communities, independently, and never merged.

Bottleneck 2 — it needs clean data

- The loss is $\|R(y) - (y - x)\|^2$. **It requires x .**
- For fluorescence microscopy, cryo-EM, MRI and astronomy a clean ground truth **does not exist and cannot be acquired** — you cannot photograph the same cell noiselessly.
- So the most valuable applications of denoising were the ones DnCNN could not serve.
- → Answered by the data branch. Presenter 2, slide 19.

Bottleneck 3 — it cannot see far enough

- Receptive field is **35×35 pixels**, chosen deliberately to match EPLL, the *smallest* effective patch size in the field. WNNM's is 361×361.
- Visible in DnCNN's own Table III: on **Barbara** it *loses* to BM3D by **0.71 dB** — the one Set12 image dominated by repetitive texture.
- The paper explains it: repetitive images *meet well with the non-local similarity prior* that a 35×35 convolution stack cannot reach.
- → Answered by the architecture branch. Presenter 2, slide 24.

A limitation visible in 2017, in the winning paper's own results table, that took until 2022 to fix. That is what an honest evolution story looks like.

2018: the field forks, and never merges

Two debts, two communities, two independent answers. Neither branch has adopted the other's advance — which is itself one of the findings of this project, and the source of the open problem we end on.

BRANCH A · BOTTLENECK 2

Change what the network is *shown*

Noise2Noise → Noise2Void → ZS-N2N → P2N. The architecture stays almost exactly where DnCNN left it; what moves is the **training target**. Each paper removes one more requirement until nothing is needed but the noisy image itself.

THE EVIDENCE THEY NEVER MERGED

P2N ships the 2018 network

In the official P2N repository the class is literally named **UNet_n2n_un**, with the 2018 Noise2Noise channel widths. Seven years, and the backbone is unchanged.
[\[measured\]](#)

BRANCH B · BOTTLENECK 3

Change what the network *is*

Restormer, NAFNet, SCUNet, MambaIR. Supervision stays exactly as DnCNN had it — paired data, same loss family. What moves is the **operator** that replaces the 3×3 convolution.

THE OPEN QUESTION

Nobody has combined them

As far as we found, **no published work trains a Transformer or state-space backbone under self-supervised RDC/DCS supervision**. We return to this as a concrete proposal on slide 40.

reading — the structural claim of this project · repository: Li-Tong-621/P2N-plus [measured]

3

Removing the Clean Image

— BRANCH A · 2018 – 2025

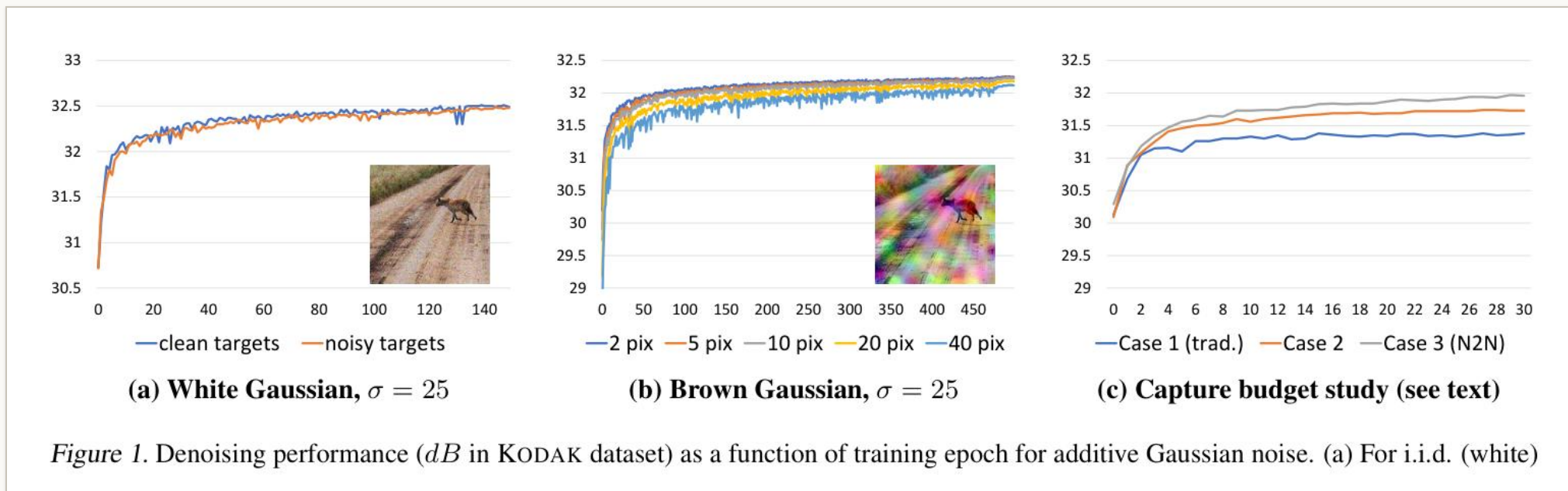
— In this part

- 01 Noise2Noise — the statistical result
- 02 Noise2Void — the blind spot
- 03 ZS-N2N and P2N — down to one image
- 04 What self-supervision costs

Four papers, each removing exactly one requirement its predecessor could not drop.

A theorem worth one conditional

Lehtinen et al., ICML 2018. The most striking ratio in this story: a substantial statistical result whose implementation change is a single branch on which tensor becomes the target.



Noise2Noise's own result: training on noisy targets (orange) tracks clean-target training (blue) almost exactly. [N2N18 · Fig. 1]

A theorem worth one conditional

the change, in essence

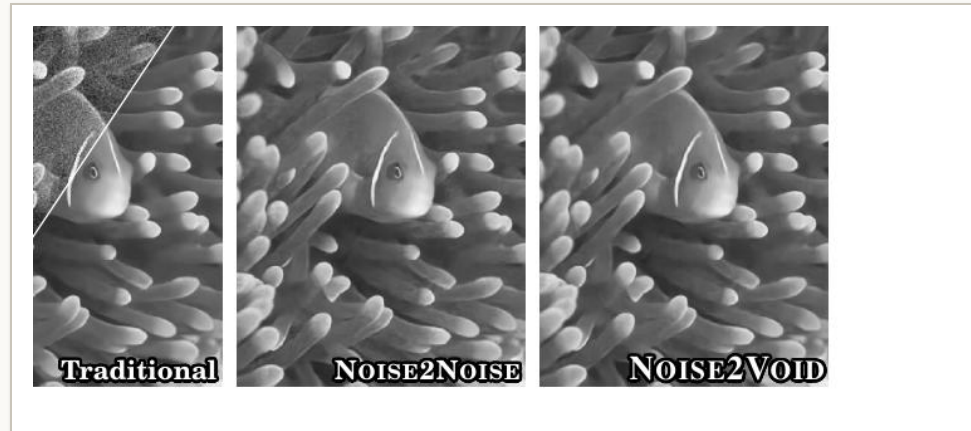
```
# BEFORE — supervised (DnCNN, 2017)
target = clean                # requires ground truth
loss   = mse(net(noisy), target)
4
# AFTER — Noise2Noise (2018)
target = second_noisy_shot    # ← the entire change
loss   = mse(net(noisy), target)
8
# network, optimiser, schedule, loss family: all unchanged
```

- 01 **Why it works.** L2 regression learns the *conditional mean* of the target. The paper states the consequence as a triviality: *on expectation, the estimate remains unchanged if we replace the targets with random numbers whose expectations match the targets.*
- 02 **So a second noisy photograph of the same scene is a valid target,** because the noise is zero-mean. Given infinite data the solution is *identical* to clean-target training; for finite data the extra variance shrinks with dataset size.
- 03 **What it does NOT need:** no likelihood model of the corruption, no density model of clean images.
- 04 **Measured cost: 0.02 dB.** Kodak/BSD300/Set14 at $\sigma=25$ — clean targets 31.63, noisy targets 31.61. The clean-data requirement was never actually necessary. [N2N18 Table 1]

N2N18 §2, Eq. (2)–(6), Table 1

N2V vs N2N — removing the second photograph

Krull et al., CVPR 2019. Noise2Noise still needs two independent shots of one scene, which rules out anything that moves and every archival image. N2V removes that too — and pays for it.



Traditional, Noise2Noise and Noise2Void training schemes side by side, with N2V's single-image blind-spot result on the right. [N2V19 · Fig. 1]

ASPECT	NOISE2NOISE (2018)	NOISE2VOID (2019)	WHY IT MATTERS
Requires	Two noisy shots per scene	One noisy image. That is all.	Applicable to any archive, any moving subject, any single acquisition
Key innovation	Corrupted target with matching mean	Blind-spot receptive field	Prediction for pixel i uses its neighbourhood <i>except pixel i</i>
Why it cannot cheat	Target is independent of input	Noise is conditionally pixel-wise independent, so neighbours carry no information about n_i	The identity solution becomes unreachable by construction

N2V19 §3.1, §3.4, §3.5, §4.1, Fig. 4 · N2N18 Table 1

N2V vs N2N — removing the second photograph

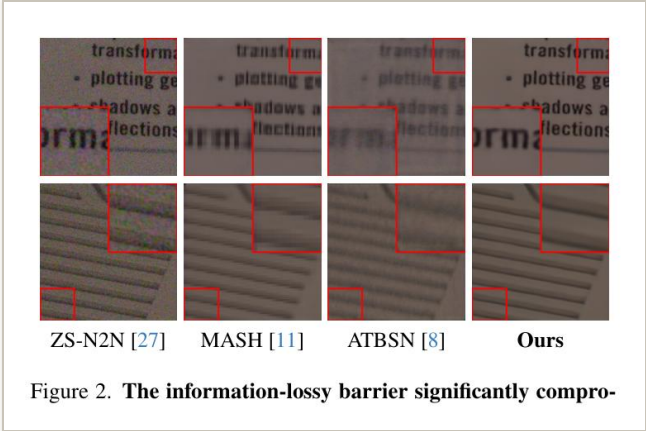
ASPECT	NOISE2NOISE (2018)	NOISE2VOID (2019)	WHY IT MATTERS
Implementation change	Swap the target tensor	Replace the centre pixel with a random neighbour; score the loss only there	A masking trick, not an architecture — any standard CNN acquires the property
Performance (BSD68 $\sigma=25$)	28.86 dB	27.71 dB	−1.15 dB, and 0.88 dB <i>below</i> BM3D — the cost is real and the paper does not hide it
Applications	Burst photography, rendering	Cryo-TEM, fluorescence microscopy, cell tracking	Domains where the paper's own figure captions the ground truth column <i>does not exist</i>

The honest framing, and the one to use if challenged. Self-supervision is not a quality technique, it is an *availability* technique. Where clean data exists, supervision wins by 1.35 dB. Where it does not, the comparison is not 27.71 against 29.23 — it is 27.71 against **nothing at all**.

N2V19 §3.1, §3.4, §3.5, §4.1, Fig. 4 · N2N18 Table 1

Down to one image, then down to no sacrifice

The branch finishes by removing the last two requirements. Compressed into one slide because the pattern is now familiar — each paper removes exactly one thing.



The information-lossy barrier made visible: ZS-N2N, MASH and ATBSN all lose fine structure that P2N's renoise-and-agree supervision keeps. [P2N25 · Fig. 2]

	ZS-N2N (2023)	P2N (2025)
Requirement removed	The training set . No corpus, no pretraining.	The information loss itself.
Key innovation	A <i>pair downsampler</i> : two fixed diagonal filters split one image into a noisy pair with similar signal and independent noise.	RDC — renoise the model's own output with + and −, giving y^+ and y^- at full resolution. DCS — train the two denoised results to agree.
Implementation change	Two fixed stride-2 convolutions; a 20 000-parameter two-layer net fitted to that one image; residual + consistency loss.	Three forward passes per step , on tensors that do not exist until the model has run once. Output compared against output, not against a target.

Down to one image, then down to no sacrifice

	ZS-N2N (2023)	P2N (2025)
Why it is not just a new loss	It is a new <i>input construction</i> .	Same: a loss scores a fixed prediction against a fixed target. P2N changes what goes in, how many passes, and which pair is compared.
Cost paid	Half the spatial resolution during training; 1–2 K gradient steps per image.	Three passes per step; still per-image test-time optimisation.
Applications	Single archival images, one-off scientific captures; ~1 min on CPU vs Self2Self's 1.2 h on GPU.	Self-supervised denoising where masking artefacts are unacceptable.

The through-line: the network never moved, the target did. Clean image (2018) → second copy (2019) → training set (2023) → information loss (2025). Four papers, four requirements removed, one U-Net throughout.

4

Replacing the Convolution

— BRANCH B · 2021 – 2024

— In this part

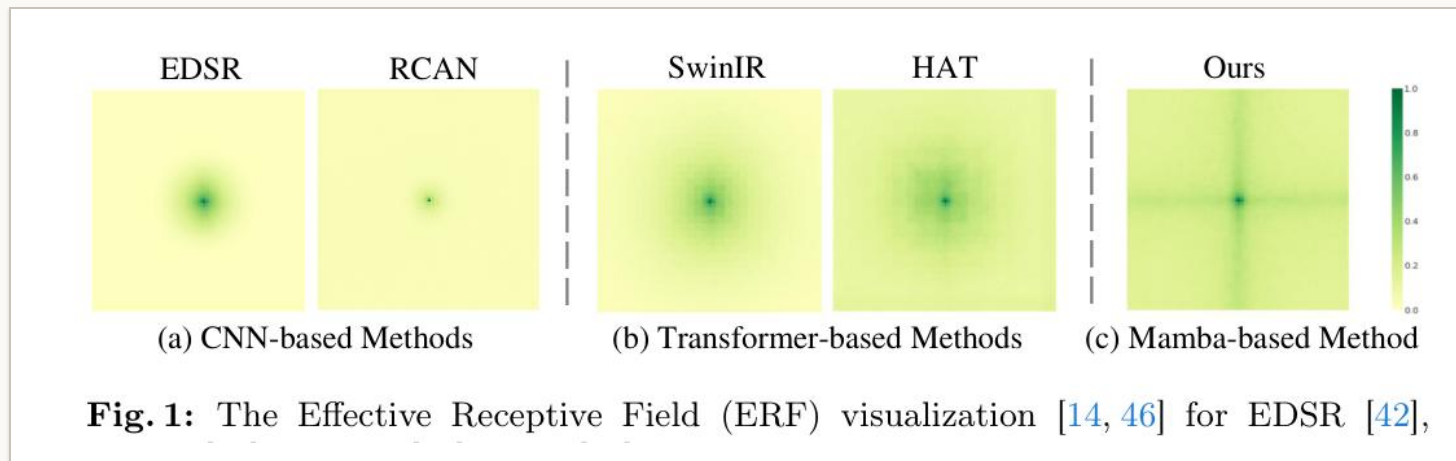
- 01 The evidence for bottleneck 3
- 02 Restormer — attention on a different axis
- 03 NAFNet — delete the mechanisms
- 04 SCUNet and MambaIR
- 05 Why there are several SOTAs

The other debt: a 35×35 receptive field, and three competing operators that fix it.

One number proves the receptive field is the problem

Identical models, identical table, identical noise level. The only thing that changes is the dataset

— and the gain changes by a factor of five.



Effective receptive field: CNNs stay tightly local, Transformers spread further, the Mamba-based backbone activates the widest field. [MambaIR24 · Fig. 1]

+0.39 dB

BSD68, $\Sigma=50$ · DNCNN → RESTORMER

Ordinary natural images, little long-range repetition. Five years of architecture research buys almost nothing.

+1.98 dB

URBAN100, $\Sigma=50$ · SAME TWO MODELS

Photographs of buildings — windows and railings repeating across hundreds of pixels. **Five times the gain**, on the dataset made of repetition.

−0.71 dB

BARBARA, $\Sigma=25$ · DNCNN VS BM3D, 2017

The same phenomenon, visible five years earlier in DnCNN's own results, on the one Set12 image dominated by repetitive texture.

Restormer22 Table 4 · DnCNN17 Table III, Table I

One number proves the receptive field is the problem

-
- 01 **What this proves.** The architecture branch did not make denoising uniformly better. It made denoising better **specifically where non-locality matters** — which is exactly the capability a 3×3 convolution stack structurally lacks.
- 02 **Why depth cannot fix it.** Receptive field grows *linearly* with depth: $(2d+1)^2$. Reaching WNNM's 361×361 the DnCNN way needs about **180 layers**. The fix had to be a different operator, not more of the same one.

Provenance: the 0.39-versus-1.98 comparison is our own analysis of published values, both taken from Restormer's Table 4 so the protocol is identical. Say *computed from Table 4* if challenged.

Restormer22 Table 4 · DnCNN17 Table III, Table I

Attention becomes affordable by changing one axis

Zamir et al., CVPR 2022. Standard self-attention costs $O(W^2H^2)$ — a 256×256 image needs a 4.3-billion-entry attention matrix, against 196 tokens for a ViT. Restormer's innovation is to attend over **channels** instead of pixels.

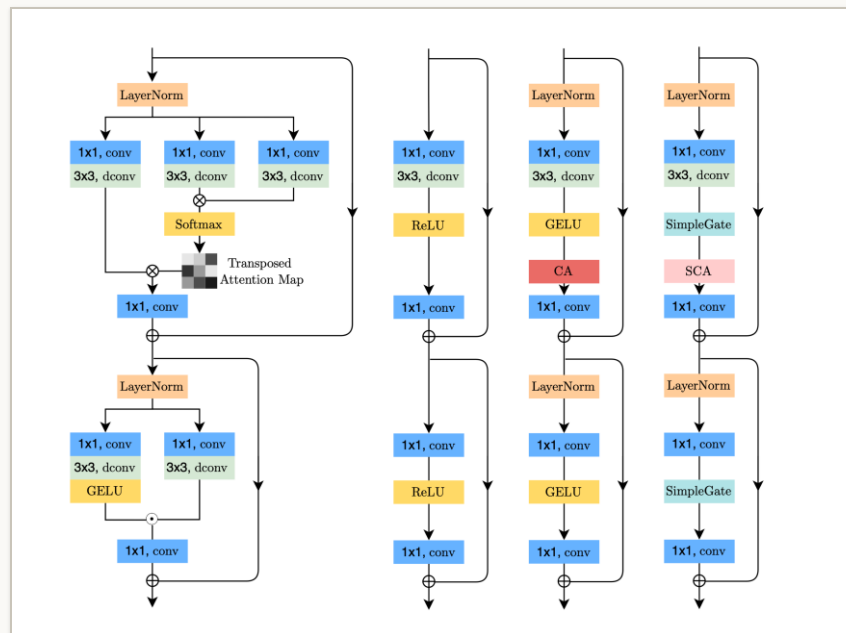
Restormer/basicsr/models/archs/restormer_arch.py — Attention.forward

```
#1 BEFORE — spatial attention (ViT / SwinIR)
#2 'b c h w -> b head (h w) c'      -> q@kT is (HW x HW)
#3 256x256 -> 65,536 x 65,536
4
#5 AFTER — Restormer MDTA
q6= rearrange(q, 'b (head c) h w -> b head c (h w)',
7         head=self.num_heads) # ← pixels on the LAST axis
k8= rearrange(k, ...); v = rearrange(v, ...)
q9= F.normalize(q, dim=-1); k = F.normalize(k, dim=-1)
attn = (q @ k.transpose(-2, -1)) * self.temperature
#1 -> (C/head x C/head). At level 1: 48 x 48, any image size
```

- 01 **One einops pattern is the entire complexity claim.** Putting pixels on the last axis makes `q @ k.transpose(-2, -1)` a C×C matrix — 48×48 whether the image is 256² or 4000×3000. Cost becomes **linear** in pixels.
- 02 **Is it really global?** Honestly: not pairwise-pixel. Every channel is computed from every pixel, so channel statistics carry image-wide information; the paper says global relationships are *implicitly* modelled and never claims otherwise.
- 03 **The depth-wise convolutions are load-bearing, not decoration.** Removing `qkv_dwconv` costs 0.15 dB on Urban100 $\sigma=50$ — more than most architecture papers gain in total. [Restormer22 Table 7]
- 04 **Why SOTA:** first method past **40 dB on both SIDD and DND**; leads Urban100 at every noise level; 3.14× fewer FLOPs and 13× faster than SwinIR. Published at CVPR 2022; how long it holds those tables is not something any source states.

The paper that asked whether any of it was necessary

Chen et al., ECCV 2022. While everyone added mechanisms, NAFNet removed them and scored higher. Its derivation is the most elegant reasoning in the modern half of this story.



The lineage NAFNet derives in one figure: the transformer block simplified step by step down to SimpleGate and SCA. [NAFNet22 · Fig. 3]

The paper that asked whether any of it was necessary

NAFNet/basicsr/models/archs/NAFNet_arch.py

```
#1 BEFORE — GELU, then a gate
#2 GELU(x) = 0.5x(1+tanh[sqrt(2/pi)(x+0.044715x^3)])
#3 CA(X) = X * sigmoid(w2 relu(w1 pool(X)))
4
#5 AFTER — SimpleGate: the gate IS the nonlinearity
def forward(self, x):
7 x1, x2 = x.chunk(2, dim=1)
8 return x1 * x2          # ← replaces GELU entirely
9
#0 AFTER — SCA: no sigmoid, no relu
self.sca = nn.Sequential(nn.AdaptiveAvgPool2d(1),
12                        nn.Conv2d(c, c, 1))
```

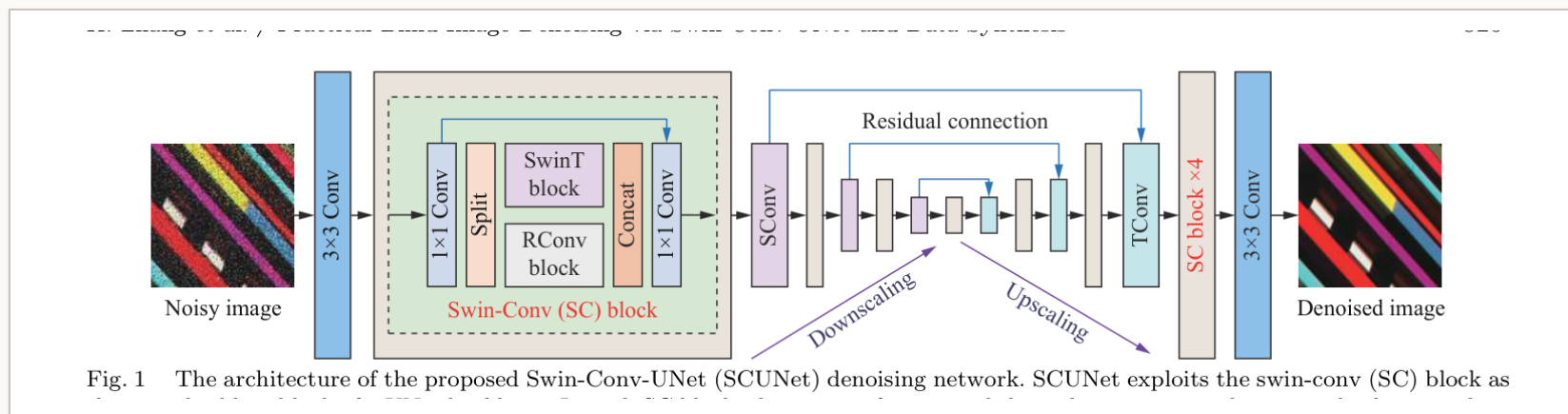
- 01 **The derivation.** GELU is a *special case* of a gated linear unit with $f, g =$ identity and $\sigma = \Phi$. But a GLU *itself contains nonlinearity and does not depend on σ* — $f(X) \odot g(X)$ is nonlinear because a product of two linear maps is not linear. So delete the activation; the gate already supplies it.
- 02 **Apply the same argument to channel attention** and both its nonlinearities vanish too. Result: **no Sigmoid, no ReLU, no GELU, no Softmax anywhere** — *the first work demonstrating that the nonlinear activation functions may not be necessary for SOTA computer vision methods.*
- 03 **Both simplifications improved the score.** SimpleGate beats GELU (+0.08), SCA beats CA (+0.03) on SIDD.
- 04 **The uncomfortable finding in their own ablation:** the largest single gain is **LayerNorm, +0.44 dB** — because it stabilises training at a 10× larger learning rate. *A large share of reported architectural progress in restoration is optimisation progress wearing a costume.*

Why SOTA: 40.30 dB on SIDD — the highest real-noise result in our comparison — at less than half the previous best's compute. Published at ECCV 2022. And by far the simplest code to maintain.

measured — megvii-research/NAFNet, verbatim · NAFNet22 §4, §5, abstract

Two more answers — one architectural, one not about architecture at all

The branch does not end with Transformers. MambaIR proposes a different operator again; SCUNet argues the operator was never the problem.



The Swin-Conv-UNet pipeline: swin-conv blocks combine local and non-local modelling on the way down and back up a UNet backbone. [SCUNet23 · Fig. 1]

	MAMBAIR (2024)	SCUNET (2023)
The claim	Attention is quadratic; windowing surrenders globality. Borrow a selective state-space model — global receptive field at linear cost.	The benchmark is the problem. A model trained on AWGN <i>performs poorly for most of real images due to noise assumption mismatch.</i>
Technical change	S6 recurrence with input-dependent B, C, Δ. Dual form: recurrent (unbounded memory) for reach, convolutional for parallel training.	Keep local <i>and</i> non-local priors in parallel — split the feature map, half to a Swin block, half to a residual conv block.

Two more answers — one architectural, one not about architecture at all

	MAMBAIR (2024)	SCUNET (2023)
Implementation change	Flatten the image to a sequence; four-directional scan ; add a local conv (fixes <i>local pixel forgetting</i>) and channel attention (fixes redundancy).	The dataloader becomes the contribution: Gaussian, Poisson, speckle, JPEG, and sensor noise synthesised through a real inverted ISP , plus resizing and double degradation.
Why SOTA / where	Best DND result, 40.04 dB — by 0.01 dB. Leads super-resolution on Manga109 ×2 by 0.41 dB over SwinIR.	Not top of any AWGN table — and does not need to be. 9.663 M parameters , smallest of the four, built to survive real photographs.
Cost / limitation	Will not import without compiled CUDA extensions. Its own changelog records that published <i>thop</i> complexity figures were wrong. [measured]	Its abstract still concedes a general-purpose blind real denoiser remains unsolved .

MambaIR24 §3.1, §3.2, Tables 1, 2, 6 · SCUNet23 abstract, §1, §3.1, §3.2 · measured

5

Why There Is No Single Winner

— SYNTHESIS

— In this part

- 01 The SOTA comparison table
- 02 Why several coexist
- 03 Applications and trade-offs
- 04 Open problems and proposals

The question this talk was built to answer, now with the numbers behind it.

Four current methods — each leads a different table

Every row names the benchmark and the metric, because a claim without those is not checkable. All quality figures are reproduced from published tables; parameters and CPU latency are our own measurements.

METHOD	LEADS ON	FIGURE	PUBLISHED	WHAT DISTINGUISHES IT	COST
NAFNet (2022)	SIDD — real sensor noise	40.30 dB	ECCV 2022	No attention, no activation functions at all; simplest code	116.0 M params · 1.90 s CPU
Restormer (2022)	Urban100 — long-range structure	28.33 dB $\sigma=50$	CVPR 2022	First past 40 dB on both SIDD and DND; channel attention at linear cost	26.1 M · 10.44 s CPU
MambaIR (2024)	DND — held-out real noise	40.04 dB	ECCV 2024	Global receptive field at linear cost via selective state-space scan	Needs compiled CUDA to import
SCUNet (2023)	Blind denoising on synthesised real-world degradation	—	Mach. Intell. Res. 2023	Competes on the training distribution , not the architecture	9.7 M · 1.69 s CPU
DnCNN (2017)	<i>Historical</i> — BSD68 Gaussian	29.23 dB	IEEE TIP 2017	The reference the whole field was built against	0.67 M · 1.02 s CPU
BM3D (2007)	<i>Zero-shot</i> — no training data	28.57 dB	IEEE TIP 2007	Requires no training data and no GPU; still the zero-shot baseline in N2N18, N2V19 and ZS-N2N23	No training · 2.85 s

No source states a SOTA window, so this deck does not assert one. The published date is verifiable; which table a method leads is verifiable from the cited comparison. For how long it held that position is not reported by any paper here, and we do not estimate it. Read the *Leads on* column: these methods are not ranked on one axis — they lead **different benchmarks**, and those benchmarks measure different capabilities.

NAFNet22 abstract · Restormer22 Tables 4, 6 · MambaIR24 Table 6 · DnCNN17 Table II · measured [Stage 2]

Four reasons the field has not converged on one method

This is the question the professor is most likely to ask, so it gets its own slide rather than a sentence.

REASON 1

The benchmarks measure different things

BSD68 measures average-case Gaussian removal and **has saturated** — 0.36 dB across five years. Urban100 measures long-range self-similarity. SIDD and DND measure real sensor noise. A method can lead one and trail another without any contradiction.

REASON 3

Deployment constraints are not quality constraints

MambaIR leads DND by 0.01 dB and **cannot be imported without compiled CUDA extensions**. SCUNet leads nothing and is a third of Restormer's size. Which one is 'best' depends entirely on what you are shipping onto what hardware.

REASON 2

The cost axes disagree with each other

Our own measurements give **three incompatible rankings** of the same models: by parameters DnCNN wins; by CPU wall-clock the P2N backbone wins; by real-noise quality NAFNet wins. NAFNet has 4.4× Restormer's parameters and runs **5.5× faster**. 'More efficient' is meaningless without naming the metric and the device.

REASON 4

Supervision availability partitions the field

Branch A and branch B answer different questions. Where clean pairs exist, supervision wins by 1.35 dB. Where they do not, the self-supervised methods are not *worse* — they are the **only** option, and the supervised SOTA is simply unavailable.

Restormer22 Table 4 · NAFNet22 · MambaIR24 · N2V19 §4.1 · measured

What you would actually deploy, and why

The honest answer to 'which is best' names the constraint first. Five constraints, five different answers, all current — which is what an unresolved research question looks like when you have to ship something.

IF YOUR CONSTRAINT IS...	DEPLOY	BECAUSE	ACCEPT
Real photos, GPU, paired data	NAFNet	40.30 dB on SIDD; best accuracy-per-millisecond we measured; simplest code to maintain	116 M parameters
Repetitive structure, max quality	Restormer	Leads Urban100 at every σ ; only method past 40 dB on both real-noise benchmarks	10.4 s per 256^2 on CPU
CPU or memory bound	SCUNet	9.7 M parameters; synthetic-degradation training survives resized and JPEG'd input	Not top of any AWGN table
No clean data exists at all	P2N / ZS-N2N	Microscopy, cryo-EM, astronomy — where ground truth cannot be acquired	Per-image test-time optimisation
No data, no GPU, no dependencies	BM3D	Seventeen years old and still beats AWGN-trained DnCNN on real noise by 1.99 dB	2.85 s, and a quality ceiling

Denoising is rarely the product. It is preprocessing for detection and segmentation, a plug-and-play prior inside deblurring and super-resolution solvers, and — per DnCNN-3 — a *template*: redefine v and the same network does super-resolution and JPEG deblocking. [SCUNet23 §1 · DnCNN17 §III-D]

What eighteen years did not solve

Each of these is grounded in a documented limitation from the papers themselves or in our own measurements — not speculation.

OPEN 1

A blind denoiser for real cameras

SCUNet's abstract states it *remains unsolved*. Neither synthetic degradation nor SIDD-trained models generalise to an **unseen camera with an unseen ISP**, and no benchmark currently measures that. DnCNN scores **23.66 dB** on SIDD — a 16 dB collapse, not a graceful degradation.

OPEN 3

Nobody agrees what 'efficient' means

Three incompatible rankings from our own measurements, and MambaIR's changelog concedes its published complexity figures were wrong. Efficiency claims across papers are close to uncheckable.

OPEN 2

Fidelity metrics reward blur

Every loss in the modern field is L1, L2 or PSNR, and all three recover a **conditional mean** — which is smoother than any plausible clean image. Noise2Noise names the mechanism itself: L2 training *results in spatial blurriness*. While leaderboards rank by PSNR, the incentive points the wrong way.

OPEN 4

The benchmarks have saturated

BSD68 $\sigma=25$ moved **0.36 dB in five years** while Urban100 moved five times as much and SIDD moved sixteen decibels. A benchmark that cannot distinguish 2017 from 2022 is no longer measuring what the field works on.

SCUNet23 abstract · Restormer22 Tables 4, 6 · N2N18 §2 · measured

Four research directions the limitations actually justify

Each is tied to a specific documented gap, states what would change, and names how it would be measured. None of these claims a paper achieved something it did not.

DIRECTION	GROUNDING IN	WHAT WOULD CHANGE	HOW YOU WOULD MEASURE IT
Cross the fork: self-supervised supervision on a modern backbone	Our own finding — no published work trains a Transformer or SSM backbone under RDC/DCS supervision; P2N still ships the 2018 U-Net	Replace UNet_n2n_un with a Restormer or NAFNet block under P2N's RDC/DCS training signal	N2V's own protocol on BSD68 $\sigma=25$ and on the cryo-TEM / CTC sets; does the 1.35 dB self-supervision gap shrink?
Cross-ISP generalisation benchmark	SCUNet's abstract: a general blind real denoiser <i>remains unsolved</i> , and no benchmark tests unseen cameras	Train on SIDD, evaluate on DND <i>and</i> a held-out camera; report the drop as the headline number	Δ PSNR between seen and unseen ISP — a metric nobody currently reports
Perceptual objective without losing fidelity	Noise2Noise names L2's <i>spatial blurriness</i> ; perceptual losses are common in SR and absent from denoising	Add an LPIPS or no-reference term to NAFNet's PSNR loss and report both axes, not one	PSNR <i>and</i> LPIPS jointly on SIDD — accept a small dB loss for a measured perceptual gain
An efficiency protocol the field can agree on	Three incompatible rankings from our measurements; MambaIR's corrected <i>thop</i> figures	Report parameters, MACs <i>and</i> wall-clock on a named device, with the counting method stated	Reproduce the table across CPU and GPU; the ratio changes are themselves the result

The first row is the one we would pursue. It falls directly out of the structural claim of this project — the two branches have never merged — and it is cheap to test, because both halves already exist as working code.

reading, grounded in P2N25, SCUNet23 abstract, N2N18 §2, MambaIR24 changelog, and our own measurements

The evolution, in one paragraph

Denoising is the cleanest case study in how computer vision replaced hand-designed priors with learned ones, because here the prior is written down explicitly and you can watch it move.

Four eras, three bottlenecks. Manual prior design ended when DnCNN made the prior a downloadable file. That created two new debts — it needed clean data, and it could only see 35 pixels — and in 2018 the field forked to pay them off separately.

Neither branch has adopted the other's advance. And because three architectures now lead three different benchmarks while none solves real-camera generalisation, the story ends in an open question rather than a winner.

— What the evidence in this talk supports.

Where to find the answer to what you are about to be asked

Kept for the Q&A. The full evidence for every claim in this talk is in the 100-slide master deck, indexed here by the question likely to prompt it.

IF ASKED...	ANSWER IN ONE LINE	MASTER DECK
Why is 0.6 dB a big deal?	Practical ceiling over BM3D was 0.3 dB, theoretical bound 0.7 dB — DnCNN took 85%	10, 37
Why predict noise, not the image?	y is closer to x than to v, so learning the image forces an identity mapping — the hard point in parameter space	33
Why do RL and BN need each other?	RL makes layer inputs Gaussian-like and content-independent — what BN's batch statistics assume	34
Why depth 17?	$RF = (2d+1)^2$ set to 35×35 to match EPLL, the smallest effective patch size in the field	35
How can noisy targets work?	L2 learns the conditional mean; the mean of a zero-mean corrupted target is the clean target	43
Where is Restormer's attention linear?	Attention.forward — the rearrange puts pixels last, so $q@k^T$ is $C \times C$ not $HW \times HW$	69
Why is NAFNet faster with more parameters?	1×1 and depth-wise convs hit optimised GEMM paths; rearrange forces tensor-layout changes	77
What did the code tell you that the papers did not?	Padding constants differ $8 \times$; SPI cuts parameters and triples latency; MambaIR needs CUDA to import	75, 77
Which would you deploy?	Name the constraint first — five constraints, five answers, all current	89
Did you verify the PSNRs yourself?	No. Every quality figure is reproduced from a named published table; only parameters and latency are ours	81, 93

The rule for the defence: do not put anything on a slide you cannot personally explain. Every claim in this talk traces to a named table in a named paper, or to a measurement whose method is stated.